Assignment 1 ShopEase – E-Commerce Application Support

Business Scenario

ShopEase is an e-commerce company that sells electronic products through its online shopping platform. The application runs on a Linux server and maintains product information, customer orders, and application logs.

As part of the application support team, you are responsible for organizing application files, verifying inventory details, investigating order failures, monitoring application logs, and maintaining backups. Complete the following activities using Linux terminal commands.

General Instructions

1. Create a separate working directory for this assignment under your Linux home directory.
2. Perform all activities using Linux terminal commands.
3. Use the sample data provided for each task.
4. Verify the results after completing each activity.
5. Do not modify or delete existing system files.
6. Do not use graphical file managers or the find command.
7. Maintain a record of the commands executed and their outputs.

Task 1: Application Directory Setup

The application team requires a directory structure to store application configuration, product inventory, customer orders, application logs, reports, and backup files.

Create the following directory structure:

```
ShopEase/
├── Application/
│   └── Config/
├── Inventory/
├── Orders/
│   ├── New/
│   └── Failed/
├── Logs/
│   └── Application/
├── Backup/
└── Reports/
```

Activities

1. Create the ShopEase directory inside your home directory.
2. Create all the required subdirectories.
3. Navigate to the ShopEase directory.
4. Display the current working directory.
5. Display the complete directory structure.
6. Create a configuration file named app.conf inside Application/Config/.
7. Add the following details to the configuration file:

```
ApplicationName=ShopEase
```

```
Environment=Production
Version=1.0
```

8. Verify the contents of the configuration file.

Task 2: Product Inventory Management

The inventory team has provided the following product details. Store these records in a CSV file so that the support team can verify product availability.

Create products.csv inside the Inventory directory.

Product Data

```
ProductID,ProductName,Category,Price,Stock,Status
P101,Laptop,Electronics,55000,15,Available
P102,Mouse,Accessories,500,100,Available
P103,Keyboard,Accessories,1200,50,Available
P104,Monitor,Electronics,12000,20,Available
P105,Printer,Electronics,8500,0,OutOfStock
P106,Webcam,Electronics,3200,0,OutOfStock
P107,Router,Networking,4500,25,Available
P108,SSD,Storage,4200,40,Available
```

Activities

1. Create products.csv using the supplied data.
2. Display the complete file content.
3. Display the first four lines.
4. Display the last three lines.
5. Count the total number of product records, excluding the header.
6. Display all products belonging to the Electronics category.
7. Identify products marked as OutOfStock.
8. Count the total number of out-of-stock products.
9. Search for keyboard without considering uppercase or lowercase letters.
10. Display the product information in a properly aligned table.
11. Save the out-of-stock product records in Reports/out_of_stock.txt.

Task 3: Customer Order Processing

The customer support team has received several complaints about failed orders. The order management application has generated the following data. Create orders.csv inside Orders/New/.

Order Data

```
OrderID,Customer,ProductID,Quantity,OrderStatus
O1001,Rahul,P101,1,Completed
O1002,Neha,P102,2,Completed
O1003,Arun,P105,1,Failed
O1004,Priya,P103,1,Completed
O1005,Kumar,P106,1,Failed
O1006,Divya,P104,2,Completed
O1007,Suresh,P107,1,Pending
```

```
O1008,Anitha,P108,1,Completed
```

Activities

1. Create the order file using the supplied data.
2. Display the first four lines.
3. Display the last three lines.
4. Count the total number of orders, excluding the header.
5. Display all completed orders.
6. Identify all failed orders.
7. Count the number of failed orders.
8. Search for the order placed by Priya.
9. Save the failed orders in Orders/Failed/failed_orders.csv.
10. Ensure the generated CSV file contains the column header.
11. Verify the contents of the generated file.

Task 4: Application Log Investigation

The application support team has received complaints about application errors and slow response times. The following log entries were captured from the ShopEase application. Create application.log inside Logs/Application/.

Application Log

```
2026-10-29 09:00:00 INFO Application started
2026-10-29 09:01:00 INFO Database connected
2026-10-29 09:02:00 ERROR Database connection timeout
2026-10-29 09:03:00 WARNING High CPU utilization
2026-10-29 09:04:00 INFO Database restored
2026-10-29 09:05:00 ERROR Product service unavailable
2026-10-29 09:06:00 INFO Customer logged in
2026-10-29 09:07:00 WARNING High memory utilization
2026-10-29 09:08:00 ERROR Order processing failed
2026-10-29 09:09:00 INFO Order service restarted
```

Activities

1. Display the complete application log.
2. Display the last five log entries.
3. Count the total number of log entries.
4. Display all ERROR messages.
5. Count the total number of errors.
6. Display all WARNING messages.
7. Display both error and warning messages.
8. Display error messages along with their line numbers.
9. Search for error without considering uppercase or lowercase letters.

10. Display one line before and one line after each error message.
11. Save all error messages in Reports/application_errors.txt.

Task 5: Application Backup and Verification

The ShopEase application is scheduled for maintenance. Before maintenance begins, the operations team requires copies of important application files.

Activities

1. Copy Inventory/products.csv into the Backup directory.
2. Rename the copied file to products_backup.csv.
3. Copy Orders/New/orders.csv into the Backup directory.
4. Copy Application/Config/app.conf into the Backup directory.
5. Display all files available in the backup directory.
6. Display backup file sizes in a human-readable format.
7. Display backup files in modification-time order.
8. Compare products.csv with products_backup.csv and verify that both contain identical data.
9. Confirm that the original files remain available in their respective directories.

Task 6: Final Application Verification

After completing the assigned activities, verify the application directory structure, data files, reports, and backups.

Activities

1. Display the complete ShopEase directory structure.
2. Verify the total number of products.
3. Verify the total number of failed customer orders.
4. Verify the total number of application errors.
5. Confirm that the following reports have been generated:

- Reports/out_of_stock.txt
- Orders/Failed/failed_orders.csv
- Reports/application_errors.txt

6. Verify that all required backup files are available.
7. Confirm that the original inventory and order files remain unchanged.

Note: All activities must be performed within the designated ShopEase directory. Existing system files and unrelated application data must not be modified or deleted.